

東京 2025 LINUX PLUMBERS CONFERENCE

TOKYO, JAPAN / DECEMBER 11-13, 2025

MM MC: Folio restructuring for huge pages (in guest_memfd)

Contact ackerleytng@google.com if you have any questions/suggestions!



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What we'd like from the community

- Thoughts on making folio restructuring orthogonal to THP and HugeTLB
- Gather other use cases for folio restructuring

Introduction to guest_memfd

- guest-first memory provider built for KVM
- The only supported memory provider for private memory for CoCo VMs

- Core principle: private guest memory must not be accessible by the host
  - Remove or disallow mappings in host page tables

guest_memfd and huge pages

- guest_memfd gets huge pages and takes ownership of them
  - HugeTLB [1]
  - Best-effort huge pages (like THP) [2]

[1] https://lore.kernel.org/all/cover.1747264138.git.ackerleytng@google.com/T/

[2] https://lore.kernel.org/all/20241212063635.712877-1-michael.roth@amd.com/T/

Background: Why restructure the folio?

- guest_memfd supports sharing memory with the host
  - aka private to shared conversion
- Shared to private conversion: guest_memfd must ensure no existing users
  - => split huge pages
    - Use struct page refcounts to track users
  - Merge folios once truncated from guest_memfd ownership

folio_split() usage similar for non-anonymous folios

buddy-allocated: file truncation or hole punching

- In process context
- folio->mapping is not NULL
- folio must have expected refcounts
- folio unmapped in split process

- Only for buddy-allocated folios
- Only split up to PMD-sized folios

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guest_memfd: private to shared conversion

- In process context
- folio->mapping is not NULL
- folio must have expected refcounts
- folio unmapped before split

- For buddy-allocated folios AND HugeTLB
- Need to split PUD-sized folios

Folio merging different

khugepaged: collapse_huge_page()

- Background periodic scanning
- Contiguous in VMAs, may not be contiguous in physical address space
- Allocate new large folio + copy contents
- Mappings removed, then restored

guest_memfd: shared to private conversion

- In process context
- Contiguous in physical address space

- Merge in-place. Preserve contents
- Mappings must be removed

Proposal: extend restructuring

- Make restructuring orthogonal to HugeTLB and THP
  - Extract folio splitting code away from page table manipulation
  - Provide folio merging functions
    - Perhaps khugepaged could use this in some cases?
- Extend restructuring beyond PMD-sized folios to PUD-sized folios
- Extend restructuring to handle HugeTLB Vmemmap Optimization (HVO)

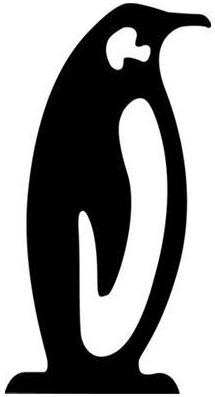
Proposal: track mergeability

- guest_memfd retains ownership of folios after split, wants to return folios *merged*
- Tracking needed:
  - Need to know if a given folio can be merged (was contiguous?)
  - Need to know initial folio size
  - Store folio fields that can't fit
  - For memory failure handling
- Are there any other use cases? Would this be useful for buddy-allocated THP?
  - Optimization to having to scan for mergeable folios?
  - Other use cases where we want to be able to merge before freeing?

Thank you!

- Please feel free to reach out to discuss anything about guest_memfd!
- I'm available in the hallways or at ackerleytng@google.com.
- guest_memfd THP: michael.roth@amd.com





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